

What changes when the rounds go: Simulation 2 against Simulation 1

LLM-free simulator, task_003, runs of 7 and 9 October 2026

The two simulations. 24 exact-Bayesian agents and one controller posting true facts on a shared board. *Simulation 1*: days and nights; every agent acts once a day; memories fade at dawn (each fact survives with probability ρ); the board is cleared each night; the controller posts b facts at once each night. *Simulation 2*: no rounds; agents and controller act at random times (controller rate λ_c); each fact has a random lifetime (on average a fraction ρ survives one time unit); readers see only the 48 most recent posts; one fact per controller message, at most 30. *Gain*: the controller’s effect on the agents’ mean belief in its target at $t = 30$, against the same episode without a controller. Compared below: Simulation 1 with $b = 1$, Simulation 2 with $\lambda_c = 1$; posts read $q = 6$ (agents), $q_c = 12$ (controller); vote gate and proof stop on.

setup	quantity	$\rho = 0.75$ (agents forget)		$\rho = 1$ (full memory)	
		Sim. 1	Sim. 2	Sim. 1	Sim. 2
nosolution	gain, truth controller (target A0)	0.42	0.29	0.38	0.38
nosolution	gain, false controller (target A2)	0.28	0.19	0.15	0.14
symmetric	gain, truth controller	0.08	0.08	0	0
symmetric	gain, false controller	0.22	0.13	0	0
symmetric	silent swarm’s belief in A0 ($q = 6 / 12$)	0.84 / 0.94	0.72 / 0.78	1	1
both	controller’s overestimate of its support	≈ 0	+0.01 to +0.15	≈ 0	+0.01 to +0.15

- 1. Asynchrony matters only through forgetting.** With full memory the two simulations agree within 0.01. When agents forget, Simulation 2 gives weaker control and a weaker consensus of the swarm on its own proof (bold entries; confirmed with fresh random draws). A reading, not yet tested: the rounds let agents hold many facts at once (dawn forgetting, then a full day of posts); with random timing, facts expire between actions. *Simulation 1 overstated both the swarm’s consensus and the controller’s power when agents forget.*
- 2. The backfire comes back by a different route.** In Simulation 1 more facts per night lowered the gain (the false controller’s gain turned negative at $b = 9$) because each all-or-nothing post had to include facts against the target. Simulation 2 posts one fact at a time, yet with full memory a *faster* controller steers *less* (false controller, nosolution: 0.20 at $\lambda_c = 0.5$, 0.05 at $\lambda_c = 8$; confirmed). Common root: every fact is true and the whole pool favours the truth, $P(A0, A1, A2) = (0.75, 0, 0.25)$; here the more different facts get out and stay, the closer beliefs move to that point (an explanation from a correlation, not yet tested directly).
- 3. Sensing goes from unbiased to biased.** The controller now overestimates its support, from about +0.014 at slow rates to +0.15 at $\lambda_c = 8$. Likely cause: the posts it reads come from recently active agents, often just after reading its own facts.
- 4. Timing effects the rounds hid.** Fast controllers overshoot and relax (symmetric, false controller at $\lambda_c = 8$: 0.26 at $t \approx 4.5$, 0.14 at $t = 30$). The effect persists ten time units after the controller stops (change at most 0.016).
- 5. The gap between truth and falsehood widens.** On symmetric the false controller loses 0.09 while the truth controller is unchanged; on nosolution both lose and the truth controller keeps its lead.
- 6. What carries over, and is therefore more trustworthy:** help with the truth replaces the agents’ own proofs; the truth controller is stronger than the false one; the vote gate saves messages at little or no cost in control; the proof stop changes nothing of size $1/24$ (one agent’s worth of belief).

Caution. λ_c stands in for b but is not the same: b facts arrive together each night with no total cap; λ_c facts arrive one at a time under a budget of 30 messages. Point 1 is the cleanest comparison; point 2 is by analogy. One world, 24 agents, probability-matching votes, 1,000 episodes per setting. Details: [analysis/task003_llm_free/simulation_2/analysis/2026-10-09_report.md](#).